

Drought Severity Forecasting with Multi-Scale Weather Features

Data Mining Final Project, 2026 Spring

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Abstract—This report describes our solution for the data mining final project. The task is to predict the next five weekly drought severity scores for each region from the most recent 91 days of meteorological observations. Our main solution is a weather-only supervised regression pipeline that converts each 91-day sequence into multi-scale tabular features and trains one tree-based regressor for each forecast horizon. The current best submitted model uses cutoff-aware regional seasonal anomaly features, weekly weather-bin summaries, drought interaction features, a filtered weather-only feature set, and a CatBoost backend. It achieves a reported Kaggle public MAE of 0.7663. Group 5. Our code is available at GitHub: [GitHub repository URL](#).

I. PROJECT SUMMARY

A. Goal

The project is a drought severity forecasting problem. For each `region_id`, the test set provides 91 consecutive daily weather observations and asks for five future weekly score predictions. The score ranges from 0 to 5, where a larger value means more severe drought. Kaggle evaluates submissions with mean absolute error (MAE), so we treat the target as an ordered regression value and submit clipped floating-point predictions instead of hard class labels.

B. High-Level Idea

Our method follows a practical tabular forecasting design. First, we convert each region’s daily weather sequence into a fixed-length feature vector. The features summarize recent weather at multiple time scales, preserve the shape of the 91-day sequence through weekly bins, compare recent conditions with each region’s own historical seasonal climatology, and add drought-specific interactions such as hot and dry conditions. Second, we train five independent tree-based regression models, one for each forecast horizon. This horizon-specific design lets week-1 and week-5 predictions learn different error patterns while keeping the implementation simple and reproducible.

The final model is intentionally weather-only. Although historical drought scores are strong persistence signals, the provided test input does not include score labels inside the 91-day forecast window. To avoid a train-test mismatch, our best reported submissions use only weather-derived features and historical regional weather climatology.

C. Related Work

This project is closely related to operational drought monitoring, tabular forecasting, and Kaggle-style weather prediction. Drought prediction usually depends on accumulated precipitation deficits, temperature stress, humidity, wind, soil or surface conditions, and strong temporal persistence. Gradient boosted decision trees are widely used for structured prediction tasks because they handle nonlinear feature interactions, mixed feature scales, missingness, and large tabular data efficiently. The Gradient Boosting method is based on Friedman, 2001 [7]. LightGBM [1] is especially suitable here because the training set contains millions of rows and the feature space is large after rolling-window expansion. The dataset also resembles the public Kaggle drought prediction dataset based on weather and soil measurements, where feature engineering from historical weather windows is a common baseline strategy [2].

II. PROPOSED METHOD

A. Problem Formulation

Let $x_{r,t}$ be the daily meteorological vector for region r on day t . For a forecast cutoff T , the input window is

$$X_{r,T} = \{x_{r,T-90}, x_{r,T-89}, \dots, x_{r,T}\}. \quad (1)$$

The goal is to predict weekly drought scores $y_{r,T+h}$ for horizons $h \in \{1, 2, 3, 4, 5\}$. We train a separate function f_h for each horizon:

$$\hat{y}_{r,T+h} = \text{clip}(f_h(\phi(X_{r,T}, r, T)), 0, 5), \quad (2)$$

where $\phi(\cdot)$ is the feature-generation function and clipping enforces the valid score range. The optimization target is MAE:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (3)$$

We choose direct multi-horizon prediction instead of a recursive forecasting scheme. In other words, the week-2 model does not consume the week-1 prediction, and the week-5 model does not depend on any earlier predicted score. This design avoids error accumulation across horizons and matches the submission format, which asks for five independent weekly outputs per region. It also makes feature importance easier to inspect because each horizon has its own trained model and its own ranking of useful weather signals.

history [0, $T - 91$]	input window [$T - 90, T$]	target $T + 7h$
baseline	weather features	weekly score

Fig. 1. Cutoff-based example construction. The cutoff T fixes the input window, future target, and historical data allowed for seasonal baselines.

B. Forecast Cutoff Construction

The forecast cutoff is the time anchor used to build each supervised example. Starting from the raw `train.csv`, we first group rows by `region_id` and convert each region into a time-ordered daily array. For a labeled target row at index t and horizon h , the cutoff is $T = t - 7h$. The input matrix is then the 91 daily observations ending at that cutoff, and the label is the weekly score at the target row. The practical pipeline is therefore:

$$\begin{aligned} \text{train.csv} &\rightarrow \text{region array} \rightarrow T \\ &\rightarrow X_{r,T} \rightarrow \phi(X_{r,T}, r, T) \rightarrow (X, y). \end{aligned} \quad (4)$$

This construction separates the selection of a cutoff from feature generation. Most features are computed only from the 91-day input window, but features that need historical context, such as regional seasonal anomalies or optional score history, must also respect the cutoff. In training, their historical baselines are built only from observations available before the 91-day input window begins. This prevents weather or score information after the forecast origin from leaking into earlier training examples.

C. Feature Generation

The raw meteorological variables are precipitation, surface pressure, humidity, temperature, dew/frost point temperature, wet bulb temperature, maximum and minimum temperature, temperature range, surface temperature, wind speed, maximum wind, minimum wind, and wind range. For each cutoff, the feature generator converts the 91-day daily matrix into one fixed-length tabular vector. The goal is to describe drought risk from multiple views: short-term shocks, accumulated deficits, region-relative seasonal deviations, physical hot-dry interactions, and the temporal shape of the recent weather sequence.

First, we compute rolling weather summaries over 7, 14, 28, 56, and 91 days. For most variables these summaries include mean, standard deviation, minimum, maximum, median, and selected percentiles. These multi-scale summaries let the model observe both short-term weather changes and slower accumulated conditions. For example, a low 7-day precipitation value may represent a brief dry spell, while a low 91-day precipitation value indicates a longer deficit. This multi-scale design follows the common drought-index idea that drought severity depends on the time scale over which precipitation and atmospheric demand are accumulated [3], [4]. This multi-scale accumulation design is based on McKee et al., 1993.

The short windows emphasize current shocks that can affect the next one or two weekly predictions, while the 56-

TABLE I
REPRESENTATIVE HIGH-IMPACT ENGINEERED FEATURES.

Feature	Function	Reference
hot_dry_w28	28-day heat stress divided by recent rainfall, capturing hot-dry atmospheric demand.	MSDI/SPEI [4], [6]
prec_region_seasonal_anom_w91	91-day precipitation deviation from the region's normal season.	SPI/SPEI [3], [4]
tmp_range_region_seasonal_anom_w91	Temperature-range deviation related to heat and moisture stress.	Review [5]
Dew-point seasonal anomaly	Air-moisture deviation from the region's normal seasonal humidity condition.	Review [5]

and 91-day windows summarize sustained conditions that are more relevant to drought persistence. Keeping both views is important because the target is weekly, but the physical drought process can accumulate over several weeks or months.

Second, precipitation receives additional event features because dry spells are not fully represented by averages alone. We compute total precipitation, dry-day count, rainy-day count, maximum consecutive dry days, recent precipitation deficit relative to the 91-day daily mean, and the fraction of dry days in the full input window. These features directly quantify rainfall absence and persistence, which are central signals in meteorological drought monitoring [3], [5].

Third, we add two forms of anomaly features. Recent-vs-long-term anomalies compare the recent 7-, 14-, and 28-day mean with the 91-day mean inside the same input window. Regional seasonal anomalies compare recent weather with the region's own historical seasonal baseline. For a variable v , window length w , region r , and cutoff T , we compute

$$a_{v,w}(r, T) = \bar{x}_{v,w}(r, T) - C_v(r, M(T, w)), \quad (5)$$

where $\bar{x}_{v,w}$ is the recent w -day mean and $C_v(r, M(T, w))$ is the region's historical month-weighted expected value for the months covered by the window. The month weighting matters because a 28-day or 91-day window can cross calendar months. Instead of assigning the entire window to one month, the expected value is a weighted combination of the monthly climatology values represented in that window. Important examples include the 91-day precipitation seasonal anomaly, the 91-day temperature-range seasonal anomaly, and dew-point seasonal anomaly. Seasonal anomalies are calculated following the SPEI index [4]. These features express whether a region is drier, hotter, or less humid than is normal for that place and season, which is more informative than a raw value when regions have very different climates [4], [5]. These drought-relevant features are supported by Mishra & Singh, 2010 [5].

For regional seasonal anomalies, the cutoff-aware implementation is important for both correctness and train-test consistency. If the climatology for an early training cutoff were computed from the full training period, the feature would silently use future weather to define what is "normal."

w1	w2	...	w11	w12	w13
oldest					newest
13-week slope: means, precipitation sums, dry days					

Fig. 2. Weekly-bin representation of the 91-day input. Week 13 is used for trend calculation but not emitted as a duplicate summary feature.

This leakage can make validation too optimistic and can create a mismatch with test-time prediction, where only the official training history is available. By using only observations before the input window, the anomaly feature has the same information boundary during training and prediction.

Fourth, we include compact drought interaction features. The main example is `hot_dry_w28`, which combines low recent precipitation with high maximum temperature over 28 days. This feature design is inspired by the MSDI index [6]. Similar evaporation-risk proxies combine high maximum temperature, high surface temperature, wind, and low humidity. These interactions are motivated by multivariate drought indices and by the fact that drought severity is driven by coupled precipitation shortage and atmospheric demand, not rainfall alone [4], [6]. They do not replace the tree ensemble’s nonlinear splits; instead, they expose sparse but physically meaningful combinations that would otherwise require several coordinated tree splits.

For example, two regions may have similar 28-day precipitation totals, but the region with higher maximum temperature and lower humidity is more likely to have strong evaporative stress. Encoding this interaction directly gives the model a domain-informed signal for the “low rainfall plus high atmospheric demand” case, instead of requiring the tree ensemble to rediscover the relation only from independent precipitation, temperature, humidity, and wind summaries.

Fifth, we preserve part of the sequence shape through weekly-bin and trend features. We divide the 91-day input into non-overlapping weekly bins. Week 1 is the oldest week and week 12 is the most recent non-duplicate emitted bin. Week 13 participates in 13-week linear trend calculation but is not emitted as a separate summary because it duplicates the existing recent 7-day features. These weekly-bin features help the model distinguish a steadily drying region from one where rainfall occurred early but not recently.

This distinction is lost if only the full-window mean is used. A window with rainfall concentrated in the oldest weeks and dry weather near the cutoff can have the same 91-day total as a window with evenly distributed rainfall, but the drought risk near the forecast origin is different. Weekly bins and 13-week slopes therefore provide a simple shape descriptor for drought development speed.

Finally, we add date and seasonality features: month, day of year, week of year, and sinusoidal day-of-year encodings. Drought risk has strong seasonal structure, so these features give the model a compact representation of the annual cycle.

```

1: for each horizon  $h \in \{1, \dots, 5\}$  do
2:   for each region time series do
3:     select weekly target rows with stride 8
4:     set cutoff  $T = t - 7h$ 
5:     extract  $X_{r,T}$  and compute  $\phi(X_{r,T}, r, T)$ 
6:     append feature row and target score
7:   end for
8:   subsample to at most 150,000 examples
9:   train tree-based regressor  $f_h$ 
10: end for

```

Fig. 3. Horizon-wise training pipeline. Each horizon has its own supervised table and its own model.

D. Training Pipeline

Training examples are generated from labeled weekly rows in `train.csv`. For each horizon, the pipeline selects historical cutoffs, extracts the previous 91 days of weather, computes features, and fits one tree-based regressor. The current main configuration uses a stride of 8 and caps each horizon at 150,000 training examples to keep training feasible. Feature matrices are stored as `float32`, and feature generation can be parallelized by region. The filtered model uses 641 weather-only features per horizon. An earlier expanded weekly-seasonal model used 871 features per horizon before filtering.

The stride is a practical sampling choice. Neighboring weekly targets from the same region produce highly overlapping 91-day windows, so using every labeled row would increase runtime and memory much more than it increases independent information. A stride of 8 keeps examples spread across time while reducing near-duplicate windows. The per-horizon cap further bounds training cost and makes feature filtering and model-backend comparisons feasible.

The prediction script loads the saved model metadata and rebuilds exactly the feature columns expected by each model. This is important because several feature filtering steps changed the feature set over time. Predictions are written in the same row order as `sample_submission.csv`.

We disable historical score features in the best submitted model even though they are strong in local validation. The reason is consistency with the final test input: the test file contains the 91-day weather window but not the recent weekly score labels inside that window. A weather-only model has weaker persistence information, but it avoids relying on a signal that is available in some validation settings and absent in the final prediction setting.

Feature extraction is also part of the method rather than only an implementation detail. The pipeline builds one horizon at a time to control peak memory usage, and parallel workers operate on per-region arrays instead of copying the full training table. This makes the expanded feature set practical enough to train while preserving the same feature-generation logic for validation, final training, and prediction.

The direct multi-horizon structure also keeps the tables simple: the week-5 model learns from weather windows and

TABLE II
MAIN TRAINING CONFIGURATION.

Item	Value
Model	Tree-based regression; CatBoost is best in model ablation
Forecast structure	One model per horizon
Horizons	1–5 weeks
Input history	91 daily weather observations
Training cap	150,000 examples per horizon
Sampling stride	8 weekly labeled rows
Best feature count	641 per horizon
Score features	Disabled in best submission
Post-processing	Clip predictions to $[0, 5]$

week-5 labels directly, rather than depending on predicted week-1 to week-4 scores. This avoids recursive error accumulation and makes feature importance interpretable horizon by horizon.

E. Filtered Feature Set

The expanded feature set is intentionally broad at first (871 features). It includes raw rolling summaries, weekly-bin summaries, recent anomalies, regional seasonal anomalies, drought interactions, and calendar features. After inspecting feature importance and inference-time behavior, we apply feature filtering to remove groups that are duplicated, constant at inference, or consistently low-gain. This step defines the final feature set used by the model.

The current filtered weather-only setup removes legacy seasonal coverage features, the inference-constant day-index feature, duplicate week-13 summaries, low-gain raw wind summaries, `prec_w56_min`, raw wet-bulb temperature summaries, and optional score-history features. Wind and wet-bulb information are not removed completely: wind seasonal anomalies, weekly bins, 13-week trends, wet-bulb weekly and seasonal features, and drought-interaction features remain. This reduces the expanded 871-column weekly-seasonal feature set to 641 weather-only features per horizon.

This filtering step also keeps prediction reproducible across feature versions. The prediction script reads the saved model metadata and rebuilds exactly the columns expected by each horizon model. Therefore, older filtered models can still be regenerated even after later feature filters are implemented in the codebase.

III. EXPERIMENTS

A. Dataset and Experimental Setup

The training set contains 12,319,040 daily rows from 2,248 regions. Each region has 5,480 training days. The test set contains 204,568 rows, corresponding to 91 days for each of the same 2,248 regions. The target score is available only on weekly labeled rows in the training set, giving 1,757,936 non-null labels. The target distribution is highly imbalanced toward

TABLE III
DATASET STATISTICS.

Statistic	Value
Training rows	12,319,040
Test rows	204,568
Regions	2,248
Training days per region	5,480
Test days per region	91
Non-null training labels	1,757,936
Prediction rows	11,240

TABLE IV
TRAINING SCORE DISTRIBUTION.

Score	Count
0	1,048,333
1	303,432
2	186,279
3	118,496
4	69,422
5	31,974

low drought severity: 1,048,333 labels are score 0, while only 31,974 labels are score 5. This imbalance encourages conservative regression predictions under MAE and makes severe drought cases harder to learn.

All experiments reported in this section were run on a local machine with an Intel Core i5-12450H CPU. Because feature extraction is CPU-bound and memory intensive, the training pipeline processes one forecast horizon at a time and uses region-level parallelism for feature generation.

Kaggle public MAE is used as the external comparison for the submitted models. We omit earlier local validation numbers because their setup used nearby score information and was not representative of the final test input, where recent score labels are unavailable.

B. Multi-Fold Time-Based Local Validation

A single time-based validation split only checks one cutoff per region, so it cannot reveal whether the local error estimate is sensitive to *which* historical period is held out. To address the multi-cutoff validation gap noted above, and to make each local validation example mirror the Kaggle test format as closely as possible (one 91-day input window producing five horizon-specific predictions per region), we construct $K = 3$ non-overlapping validation folds per region.

For region r , let $t_1 < t_2 < \dots < t_{m_r}$ denote the day indices that carry a non-null weekly score, ordered from oldest to most recent. With $w = 5$ labeled weeks per fold, fold $k \in \{0, 1, \dots, K - 1\}$ (where $k = 0$ is the most recent fold and is closest in time to the Kaggle test split) is anchored at the cutoff

$$T_r^{(k)} = t_{m_r - (k+1)w + 1} - 7. \quad (6)$$

Fold k contributes a single 91-day input window $X_{r, T_r^{(k)}}$, which is reused to generate one validation example per hori-

training cutoffs	fold 2	fold 1	fold 0
cutoff $< T_2 - 7g$	$[T_2, T_2+35]$	$[T_1, T_1+35]$	$[T_0, T_0+35]$
oldest $\rightarrow$ newest	(fold 0 closest to the Kaggle test split)		

Fig. 4. Layout of the $K = 3$ non-overlapping validation folds. Here $T_k \equiv T_r^{(k)}$ for brevity. Each fold k reuses one 91-day input window ending at T_k to predict all five horizons $T_k+7, \dots, T_k+35$. Training cutoffs are restricted to targets earlier than fold 2’s window, optionally shifted further back by a gap of g weeks.

zon:

$$\left(\phi(X_{r, T_r^{(k)}}(r, T_r^{(k)}), y_{r, T_r^{(k)}+7h}\right), \quad h = 1, \dots, 5. \quad (7)$$

In other words, each fold behaves exactly like one test-time region: a fixed 91-day weather window that is scored against five future weekly labels, one per horizon.

Training examples for every horizon are drawn only from cutoffs whose horizon- h target lies strictly before fold 2’s window, i.e. before t_{m_r-Kw+1} , so that no validation label can leak into training. We also introduce an optional gap of g weeks (`train_gap_weeks`) that removes the g weeks immediately preceding fold 2 from the training pool, emulating the roughly 60-week separation between Kaggle’s visible training history and its hidden test targets. All other settings (the 641-feature filtered weather-only feature set, stride 8, a cap of 150,000 training examples per horizon, and the LightGBM backend) are unchanged from Section II-D so that the multi-fold result is directly comparable to the single-split numbers reported earlier.

C. Hyperparameter and Configuration Tuning

We systematically explored the effects of three key training configurations: stride (sampling interval for examples), maximum training examples per horizon, and the inclusion of historical score features. The stride parameter controls the temporal density of training examples sampled from the full time series; larger strides reduce training size but may improve generalization by avoiding temporal autocorrelation. The max training examples cap controls total training set size per horizon. Finally, the no-score variant trains on weather-only features without using historical drought scores, which is critical for the final submission because Kaggle’s test input lacks score labels within the 91-day window. Table V shows the Kaggle public MAE results across ten configurations.

The results clearly show that excluding historical score features significantly improves performance, confirming the train-test mismatch concern discussed in Section II. Among the no-score configurations, 150k with stride 8 achieved the best public MAE of 0.8841. This combination provides sufficient training data (150,000 examples per horizon) while maintaining good generalization through moderate sampling (stride 8). All subsequent tuning and feature experiments use this 150k-s8-no-score configuration.

D. Feature Ablation

We perform feature ablation on the 641-feature filtered LightGBM setup. Each experiment removes one high-gain

TABLE V
HYPERPARAMETER AND CONFIGURATION TUNING.

Configuration	Public MAE
150k stride 8, no score	0.8841
100k stride 8, no score	0.8870
80k stride 4, no score	0.9016
80k stride 8, no score	0.9017
150k stride 4, no score	0.9065
80k stride 4, with score	0.9175
80k stride 8, with score	0.9254
150k stride 8, with score	0.9333
150k stride 4, with score	0.9434
100k stride 8, with score	1.0609

TABLE VI
FEATURE ABLATION ON THE 641-FEATURE FILTERED SETUP. LOWER IS BETTER.

Setting	Public MAE
Original 641-feature model	0.7745
w/o hot_dry_w28	0.7850
w/o prec_region_seasonal_anom_w91	0.7770
w/o tmp_range_region_seasonal_anom_w91	0.7707

feature group while keeping the same training size, stride, weather-only setting, and prediction pipeline. The original 641-feature LightGBM submission obtains a public MAE of 0.7745.

Removing `hot_dry_w28` causes the largest degradation, increasing MAE from 0.7745 to 0.7850. This confirms that the hot-dry interaction is a key feature for the current weather-only model: it combines precipitation shortage and heat stress into one direct drought-risk signal, so the tree model loses an important cross-variable relation when it is removed.

Removing the 91-day precipitation seasonal anomaly also hurts performance, raising MAE to 0.7770. This is expected because long-term precipitation deficit relative to a region’s normal season is one of the most direct indicators of meteorological drought. Without this feature, the model can still use raw precipitation summaries, but it has less information about whether the same rainfall amount is unusually low for that region and time of year.

In contrast, removing the 91-day temperature-range seasonal anomaly improves the public MAE to 0.7707. Although this feature has high gain importance, the removal result suggests that it may be redundant with other temperature, humidity, seasonality, or interaction features, or that it overfits the public split in the current LightGBM model. This shows why gain importance should be interpreted together with removal-based ablation.

E. Model Ablation

To isolate the effect of the learning algorithm, we keep the same 641-feature filtered table and replace only the model backend. All model backbones use the same weather-only

TABLE VII
MODEL ABLATION USING THE SAME 641-FEATURE TABLE. LOWER IS BETTER.

Backbone	Public MAE
LightGBM	0.7745
XGBoost	0.7723
Random Forest	0.8012
CatBoost	0.7663

setup, stride, training cap, and horizon-wise prediction structure.

XGBoost slightly improves over LightGBM, reducing public MAE from 0.7745 to 0.7723. Since both are gradient-boosted tree models, the small gap suggests that the engineered feature table is already strong and that backend-specific tree growth and regularization choices provide only a moderate additional gain.

Random Forest performs worse than the boosting-based models, with a public MAE of 0.8012. For this model, the number of trees was set to 80, the maximum depth to 24, the minimum samples per leaf to 10, the fraction of samples per tree to 0.5, and the number of features considered at each split to the square root of the total features. This result suggests that the ensemble benefits here come more from boosted sequential correction than from a pure bagging-style forest.

CatBoost achieves the best public score among the tested backbones, reducing MAE to 0.7663. A likely reason is that CatBoost’s ordered boosting and stronger default regularization are more robust to the large, correlated feature table created by rolling summaries, weekly bins, and seasonal anomalies. This result indicates that the proposed feature representation is not tied to one tree-boosting implementation, and that the final performance benefits from selecting the strongest backend after feature engineering is fixed.

F. Current Limitations and Remaining Work

The strongest current external result is the Kaggle public score of 0.7663, but public leaderboard feedback should not be the only validation signal. The latest filtered feature versions still need multi-cutoff local validation. This is important because a single public score can be noisy and may encourage overfitting to the public split.

Runtime is another limitation. Feature extraction is the bottleneck because each horizon requires many 91-day windows and hundreds of engineered features. The implementation already reduces memory pressure by building one horizon at a time, using `float32` matrices, and parallelizing by region. Future work should add stable intermediate caching only after the feature set stops changing.

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